SEMESTER 6

INFORMATION TECHNOLOGY

SEMESTER S6

CRYPTOGRAPHY AND NETWORK SECURITY

Course Code	PCITT601	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To explain the basics of Security and mathematical foundations of cryptography
2. To describe Cryptography fundamentals and traditional ciphers
3. To interpret the standard symmetric and asymmetric cryptosystems.
4. To create an awareness for the design of various cryptographic primitives
5. To analyze different types of attacks on various cryptosystems.
6. To create an understanding of the modern applications of cryptography.

SYLLABUS

Module No.	Syllabus Description	Contact Hours (44 Hrs)
1	Introduction to Security:- Security Goals – Security services (Confidentiality, Integrity, Authentication, Non-repudiation, Access control) – Security Mechanisms (Encipherment, Data Integrity, Digital Signature, Authentication Exchange, Traffic Padding, Routing Control, Notarization, Access control), Security Principles. Basics of Algebra and Number Theory:- Integer Arithmetic- Modular Arithmetic- Algebraic structures – Prime Numbers - Fermat's and Euler's Theorem – Factorization - Chinese Remainder Theorem - Linear and Quadratic Congruence - Discrete Logarithms.	10
2	Introduction to Cryptography:- Kerckhoff's Principle -Classification of Cryptosystems- Cryptanalytic attacks- Cipher Properties (Confusion, Diffusion). Traditional Secret Key Ciphers:- Substitution Ciphers – (Mono alphabetic ciphers-Ceaser, Affine, Poly alphabetic ciphers-Autokey, Playfair,	12

	Vigenere and Hill)- Transposition Ciphers – (Rail Fence, Keyed transposition ciphers) - Stream and Block Ciphers. Modern Secret Key Ciphers:- Substitution Box-Permutation Box-Product Ciphers, Data Encryption Standard (DES) (Fiestel and Non-Fiestel Ciphers, Structure of DES, DES Attacks, 2-DES, 3-DES) - Advanced Encryption Standard (AES) (Structure, Analysis)	
3	Cryptographic Hash Functions:– Properties - Secure Hash Algorithm- (SHA-512 Logic, Round Function), Message Authentication Code (MAC). Public Key Cryptosystems (PKC):- Types of PKC– Trapdoor - one way functions -RSA Cryptosystem (Integer Factorisation Trapdoor, Key Generation, Encryption, Decryption) - El Gamal Cryptosystem (Discrete Logarithm Trapdoor, Key Generation, Encryption, Decryption) - Diffie-Hellman Key Exchange Protocol, Man in the Middle attack on Diffie-Hellman Protocol. Digital Signature:- Signing – Verification - Digital signature forgery (Existential forgery, Selective forgery, Universal forgery) - RSA Digital Signature Scheme - ElGamal Signature Scheme – Elliptic Curve Digital Signature Scheme(ECDSS)	**11**
4	Network Security:- Electronic Mail Security -Pretty Good Privacy- PGP message format - Transmission and Reception of PGP Messages, S/MIME. IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload- Intruders, Intrusion Detection, Distributed Denial of Service attacks- Secure Electronic Transaction – Payment Processing - Dual Signature, Firewalls - Firewall Design Principles. Application of Cryptography:- Blockchain fundamentals(Blockchain defined, Blockchain architecture, Generic elements of a blockchain, Benefits, features, and limitations of blockchain, Types of blockchain) - Secure Multiparty Computation(Outsourced Computation, Multi-Party Computation, Applications).	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

<table>
<tr><td colspan="3">In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions</td></tr>
<tr><th>Part A</th><th>Part B</th><th>Total</th></tr>
<tr><td>• 2 Questions from each module.
• Total of 8 Questions, each carrying 3 marks

(8x3 =24marks)</td><td>• Each question carries 9 marks.
• Two questions will be given from each module, out of which 1 question should be answered.
• Each question can have a maximum of 3 sub divisions.
(4x9 = 36 marks)</td><td>60</td></tr>
</table>

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe the various network security aspects and apply number theory concepts in solving cryptographic problems.	**K3**
CO2	Illustrate the process of enciphering and deciphering of classical cryptosystems and modern symmetric key cryptosystems.	**K3**
CO3	Apply the principles of asymmetric key cryptosystems and digital signature.	**K3**
CO4	Discuss various protocols to ensure Email Security and Network Security.	**K2**
CO5	Explain the various applications of cryptography	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2								3
CO2	3	3	3	2		1						3
CO3	3	3	3	2		1						3
CO4	3	2	2	2		1						3
CO5	2	2	2	2		1						3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cryptography and Network Security	William Stallings	Pearson Education	7th Edition, 2017
2	Cryptography & Network Security, Second Edition	Behrouz A. Forouzan and Debdeep Mukhopadhyay	Tata McGraw Hill, New Delhi,	3rd Edition, 2015
3	Mastering BlockChain	Imran Bashir	Packt	3rd Edition 2020
4	A Pragmatic Introduction to Secure Multi-Party Computation	David Evans; Vladimir Kolesnikov; Mike Rosulek	NOW Publishers	2018

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cryptography and Network Security	Atul Kahate	Tata McGraw Hill	3rd edition 2017
2	Network Security and Cryptography	Bernard Menezes	Cengage Learning India	2011
3	Applied Cryptography: Protocols, Algorthms, and Source Code in C”, Second Edition, , 2001.	Bruce Schneier	John Wiley and Sons Inc	2001

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://archive.nptel.ac.in/courses/106/105/106105162/
2	https://onlinecourses.swayam2.ac.in/cec24_cs16/preview

SEMESTER S6

ADVANCED ARTIFICIAL INTELLIGENCE

Course Code	**PCITT602**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide a comprehensive overview of modern AI technologies, focusing on both traditional methods and recent advancements.
2. To learn methods and practices for making AI decisions transparent and understandable.
3. To provide a comprehensive understanding of the principles, practices, and implications of responsible AI.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Foundations of Modern Artificial Intelligence (AI): Introduction to AI- Historical Overview and Definitions, AI System Categories- Systems that think like humans, act like humans, think rationally and act rationally, AI Agents and Environment- Simple Reflex, Model Based, Goal Based, and Utility Based Agents. Traditional AI Techniques- Outline of Knowledge based Systems and Expert Systems. Modern AI Techniques- Outline of Federated Learning, Reinforcement Learning and Transfer Learning.	**7**
2	AI Search Strategies: Uninformed Search Strategies-Depth First Search, Breadth First Search, Depth Limited Search, Iterative Deepening Depth-First Search, Informed search strategies- Hill Climbing, Best First Search, A* algorithm, AO* algorithm, Game Playing- Minimax algorithm, Alpha Beta Pruning, Constraint Satisfaction Problem- Backtracking, Constraint Propagation, Comparison between Informed and Uninformed Search Strategies.	**10**

3	Explainable AI (XAI): Introduction to Explainable AI- Importance and challenges of XAI, Overview of XAI techniques, Interpretable Models- Decision trees, linear models, and rule-based models, Trade-offs between interpretability and accuracy, Post-Hoc Explanation Techniques- Local Interpretable Model-agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP), Counterfactual explanation, Evaluating Explanations- Metrics for evaluating the quality and usefulness of explanations, User studies and feedback mechanisms, Regulatory and Ethical Considerations- Compliance with regulations like GDPR and the AI Act, Balancing transparency with privacy.	**11**
4	Responsible AI, Societal Impact, and Future Directions: Responsible AI Practices- Principles of responsible AI (fairness, accountability, transparency), Designing and auditing AI systems for ethical compliance, Societal Impact of AI- AI's role in various sectors (healthcare, finance, transportation), Privacy concerns and data security, Future Directions and Emerging Trends- Advances in AI research: neuromorphic computing, quantum AI, Potential societal transformations and ethical considerations	**8**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the evolution of AI from its historical roots to contemporary techniques, including traditional methods and modern approaches.	**K2**
CO2	Implement various search strategies used in artificial intelligence.	**K3**
CO3	Apply various XAI techniques and assess the trade-offs between interpretability and accuracy in AI systems.	**K3**
CO4	Apply principles of responsible AI, assess AI's societal impact across different sectors, and understand the regulatory and ethical considerations influencing AI development and deployment.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	2	1	-	-	2	-	-	2
CO2	3	3	3	3	2	1	-	-	2	-	-	2
CO3	3	2	3	3	2	2	-	-	2	-	-	2
CO4	2	2	2	2	3	2	-	2	2	-	2	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	Pearson	2016
2	An introduction to machine learning interpretability	Hall, Patrick, and Navdeep Gil	O'Reilly Media	2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Artificial Intelligence: A Guide for Thinking Humans	Melanie Mitchell		2019
2	AI Ethics	Coeckelbergh M.	MIT Press	2020

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://www.javatpoint.com/artificial-intelligence-ai
2	https://www.solulab.com/ai-deep-learning-techniques/
3	https://onlinecourses.nptel.ac.in/noc22_cs56/preview
4	https://onlinecourses.nptel.ac.in/noc24_cs132/preview

SEMESTER S6
COMPILER DESIGN

Course Code	**PEITT631**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT303 - Data structures PEITT525 - Formal Languages and Automata Theory	**Course Type**	Theory

Course Objectives:

1. To provide students with a foundational knowledge of compiler design, including the need for compilers, their phases, and the tools used in their construction.

2. To enable students to design and implement lexical and syntax analyzers using various parsing methods, including both top-down and bottom-up approaches.

3. To equip students with the skills to generate intermediate code, optimize code, and develop a basic code generator.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Fundamentals:** Need of compiler, Analysis of the source program, Phases of a compiler, grouping of phases, compiler writing tools, Cross compiler. **Lexical Analyser:** The role of Lexical Analyzer, Review of Finite Automata, Design of Lexical Analyser using DFAs, Specification of Tokens using Regular Expressions.	**9**
2	**Syntax Analyser:** Review context-free grammars, Parse Trees, Ambiguous Grammars. Top – Down Parsing: Recursive Descent Parser, Predictive Parser-LL(1). Bottom-Up Parsing: LR(0) parser, SLR(1) parser, CLR(1) parser, LALR parser.	**9**
3	**Semantic Analyser:** Syntax directed definitions: S-attributed definitions, L-attributed definitions. Bottom- up evaluation of S- attributed definitions, Top-down translation, Bottom-up evaluation of inherited attributes. **Intermediate Code Generation (ICG):** Need of ICG, Different representation of intermediate code, intermediate code for assignment statement.	**9**
4	**Code Optimization:** Principal sources of optimization, Optimization of basic blocks. **Code Generation:** Issues in the design of a code generator, A simple code generator.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe the phases of a compiler, their functions, and the tools utilized in compiler construction.	**K2**
CO2	Design and implement a lexical analyzer using deterministic finite automata (DFAs) and regular expressions.	**K3**
CO3	Apply various parsing techniques, including top-down (recursive descent and LL(1)) and bottom-up (LR(0), SLR(1), CLR(1), and LALR) methods, and construct parse trees.	**K3**
CO4	Develop test automation strategies and utilize automation and testing tools effectively to test object-oriented software and web-based systems, while applying debugging techniques to track and resolve bugs in various environments	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	2	-	2	-	-	-	-	-	-	-
CO3	3	3	3	-	2	-	-	-	-	-	-	-
CO4	2	2	2	3	3	-	-	-	-	-	-	-

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Compilers Principles, Techniques and Tools	Alfred Aho, Ravi Sethi and Jeffrey D Ullman	Pearson Education	Updated Second Edition, 2023

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Compiler Construction: Principles and Practice	Kenneth C. Louden	PWS Publishing Company	Second Edition, 1997
2	Lex &Yacc	Levine J.R, Mason T, Brown D	OReilly Associates	First edition, 1992
3	Compiler Design in C	Allen I. Holub	Prentice Hall	First edition, 2003

Video Links (NPTEL, SWAYAM…)

No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105190/
2	https://archive.nptel.ac.in/courses/106/105/106105190/
3	https://nptel.ac.in/courses/106108113
4	https://nptel.ac.in/courses/106108113

SEMESTER S6

METAHEURISTIC OPTIMIZATION

Course Code	PEITT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT303 Data structures	Course Type	Theory

Course Objectives:

1. To provide a structured framework for students to grasp both theoretical and practical aspects of metaheuristics.
2. To enable the students to effectively use and evaluate metaheuristics in various optimization scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours (36 Hrs)
1	Fundamentals of Metaheuristics: Classical Optimization Models, Optimization Methods-Exact Methods and Approximate Methods, Metaheuristics Concepts–Representation, Objective Function, Constraint Handling, Parameter Tuning and Performance Analysis.	9
2	Single Solution Based Metaheuristics: Local Search- Neighbor Selection and Escaping from Local Optima, Simulated Annealing- Move Acceptance and Cooling Schedule, Tabu Search- Types of memory, Iterated Local Search-Perturbation Method and Acceptance Criteria, Variable Neighbourhood Search- Variable Neighborhood Descent and General Variable Neighborhood Search.	9
3	Ant Colony Optimization (ACO): Inspiration from Nature-Ant Foraging Behavior, Ant System, Basic Principles of ACO- Pheromone Deposition, Pheromone Updating Rules, Construction of Solutions and Pheromone Evaporation, Ant System Variants-Elitist Ant System, Rank Based Ant	9

	System, Max Min Ant System, Application of ACO to the Travelling Salesman Problem.	
4	Particle Swarm Optimization (PSO): Inspiration from Nature- Social Behavior of Bird Flock, Basic Models of PSO- Local Best PSO and Global Best PSO, gbest versus lbest PSO, Velocity Update Equations, Position Update Equations, Velocity Clamping, Inertia Weight, Constriction Coefficients, Synchronous and Asynchronous Updates, Velocity Models, Basic PSO Parameters, Guaranteed Convergence PSO, Problem Formulation of PSO algorithm, Advantages and Disadvantages of PSO.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain classical optimization models, optimization methods and explain the main concepts of metaheuristics.	**K2**
CO2	Apply single solution based metaheuristics to solve optimization problems.	**K3**
CO3	Analyze the optimization problem and apply ant colony optimization algorithm to solve it.	**K4**
CO4	Analyze real world problems and solve them using particle swarm optimization algorithm.	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	-	-	-	-	2
CO2	2	3	3	2	2	2	-	-	2	-	2	2
CO3	2	3	3	3	2	2	-	-	2	-	2	2
CO4	2	3	3	3	2	2	-	-	2	-	2	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Metaheuristics: from design to implementation	Talbi, El–Ghazali	John Wiley & Sons	Vol. 74, 2009
2	Essentials of Metaheuristics	Luke, Sean	Lulu	Second edition, 2013
3	Handbook of metaheuristics	Gendreau, Michel, and Jean-Yves Potvin	New York: Springer	Vol. 2, 2010
4	Handbook of approximation algorithms and metaheuristics	Gonzalez, Teofilo F	CRC Press	2007

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the**	**Edition and**

			Publisher	Year
1	Computational intelligence: an introduction	Engelbrecht, A. P	John Wiley & Sons	2007
2	Ant Colony optimization	Marco Dorigo and Thomas Stutzle	MIT Press	2004
3	Swarm Intelligence: From Natural to Artificial Systems	Eric Bonabeau, Marco Dorigo, Guy Theraulaz	Oxford University Press	2000

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://nptel.ac.in/courses/105103210
2	https://nptel.ac.in/courses/110105096
3	https://nptel.ac.in/courses/103103164

SEMESTER S6

SOFTWARE PROJECT MANAGEMENT

Course Code	PEITT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT503 - Software Engineering	Course Type	Theory

Course Objectives:

1. To provide an in-depth understanding of the software project management process, including its key concepts, methodologies, and frameworks.
2. To equip students with the skills required to evaluate, plan, and estimate software projects, ensuring effective management of project portfolios and individual projects.
3. To develop students' ability to apply principles of activity planning, risk management, and resource allocation to ensure efficient sequencing, scheduling, and management of project activities and resources.
4. To enhance students' knowledge of monitoring and control processes, contract management, people management, and teamwork, enabling them to support successful project execution and delivery.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Software Project Management**: Introduction, Importance, Activities, Plans, methods and methodologies, Categorization, Project Charter, Stake holders, Setting Objectives, **Management**: Management Control, Project Management Life Cycle, Traditional vs Modern project management	9
2	**Project Evaluation:** Project portfolio management- Evaluation of individual projects- Cost benefit evaluation techniques- Risk evaluation- Programme Management- Managing allocation of resources-Strategic Programme Management-Creating a Programme- Aids to Programme Management-	9

	Benefits Management. **Project Planning**: Step wise Project Planning **Software Estimation:** Basis for software estimation- Software Effort estimation techniques- Bottom-up and Top-down estimation- Function Point Analysis- COCOMO II. Cost Estimation- Staffing Pattern- Schedule compression.	
3	**Activity Planning:** Objectives- Project Schedules- Projects and Activities- Sequencing and Scheduling Activities- Network Planning Models- Forward Pass- Backward pass- Identifying Critical Path – Activity Float- Shortening the Project Duration- Identifying Critical Activities- Activity-on-arrow networks. **Risk Management:** Risk- Categories of Risk- Risk Management Approaches-Risk Identification- Risk Assessment- Risk Planning- Risk management- Risk Evaluation- PERT, Monte Carlo Simulation, Critical Chain. **Resource Allocation:** Nature of Resources- Identifying and Scheduling Resources- Creating Critical Paths- Cost Schedule- Scheduling sequence.	**9**
4	**Monitoring and Control:** Creating the framework- Collecting data- Review - Visualizing Progress- Cost Monitoring- Earned Value Analysis- Prioritizing Monitoring-Getting the project back to target- Change control- Software Configuration Management. **Managing Contracts**- Types of Contract-Stages in Contract Placement- Contract Management **Managing People:** Organizational Behaviour- Selecting the right Person- Motivation- The Oldham-Hackman Job Characteristics Model-Stress- Stress Management- Health and Safety- Ethical and Professional Concerns **Working in Teams**- Becoming a Team- Decision Making- Organization and Team Structures- Coordination Dependencies- Dispersed and Virtual Teams – Communication Genres- Communication Plans- Leadership.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of software project management	**K2**
CO2	Apply project evaluation, planning, and software estimation techniques to effectively manage project portfolios, individual projects ensuring optimal resource allocation and cost estimation	**K3**
CO3	Describe the principles of activity planning, risk management and resource allocation to effectively sequence, schedule and manage project activities and resources	**K2**
CO4	Explain the principles of monitoring and control, contract management, people management, and team work to effectively support project execution and delivery	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	2	-	-	-	2	3	2
CO2	3	2	-	-	-	-	-	-	2	2	3	2
CO3	3	-	-	-	2	-	-	-	2	3	3	2
CO4	3	-	-	-	-	-	-	-	3	2	3	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Software Project Management	Bob Hughes, Mike Cotterell, Rajib Mall	Mc Graw Hill India	6th edition, 2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Software Project Management in Practice	Pankaj Jalote	Pearson	1st edition, 2015
2	Software Engineering	Ian Somerville	Pearson	10th edition,2017
3	Software Engineering : A Practitioner's Approach	Roger S Pressman, Bruce R Maxim	Tata McGraw Hill	8th edition, 2019

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_mg78/preview

SEMESTER S6

QUANTUM COMPUTING

Course Code	PEITT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GAPHT121: Physics for Information Science	Course Type	Theory

Course Objectives:

1. Understand the fundamentals of quantum computing and quantum information.
2. Explore and implement quantum algorithms.
3. Learn and apply principles of linear algebra and quantum mechanics.
4. Design and analyze quantum circuits and universal quantum gates.
5. Explore advanced topics in quantum computation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Fundamental Concepts:** History of quantum computation and quantum information, Quantum bits, Quantum computation – Single qubit gates-Multiple qubit gates-Quantum Circuits-Qubit copying circuit-Example: Bell states. Quantum algorithms-Classical computations on a quantum computer-Quantum parallelism-Deutsch's algorithm-The Deutsch–Jozsa algorithm.	9
2	**Introduction to quantum mechanisms**: Linear algebra, Bases and linear independence- Linear operators and matrices- The Pauli matrices- Inner products-Eigenvectors and eigenvalues-Adjoints and Hermitian operators-Tensor products-Operator functions.	9
3	**The Postulates of Quantum Mechanics:** State space-Evolution-Quantum measurement-Distinguishing quantum states-Projective measurements-POVM measurements-Phase- Composite systems.	9
4	**Quantum Computation:** Quantum circuits-Universal quantum gates-Simulation of quantum systems. The quantum Fourier transform- Phase estimation-Application: order-finding-Quantum search algorithms-Quantum search as a quantum simulation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Discuss the historical evolution, foundational concepts, and essential algorithms in quantum computation and quantum information, encompassing qubits, gates, and circuits.	**K2**
CO2	Describe the fundamental concepts of quantum mechanics, including linear algebra, bases, linear independence, operators, matrices, and key elements like the Pauli matrices and eigenvalues.	**K2**
CO3	Discuss the postulates of quantum mechanics, including state space, evolution, quantum measurement, distinguishing quantum states, projective measurements, POVM measurements, phase, and composite systems.	**K2**
CO4	Apply knowledge of quantum computation to design quantum circuits using universal quantum gates, simulate quantum systems, perform the quantum Fourier transform, estimate phases, implement order-finding applications, and utilize quantum search algorithms for quantum simulation.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2				3					
CO2	3	3	2									
CO3	3	2	3									
CO4	3	3	2									2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Quantum Computation and Quantum Information	Michael A. Nielsen & Isaac L. Chuang	Cambridge University Press	10^{th} Edition, 2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	An Introduction to Quantum Computing	Kaye P, Laflamme R	Oxford University Press	2007
2	Quantum Computer Science: An Introduction	Mermin N.D.	Cambridge University Press	2007

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://nptel.ac.in/courses/106106232
2	https://onlinecourses.nptel.ac.in/noc21_cs103/preview

SEMESTER S6

DATA ANALYTICS

Course Code	**PEITT635**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PBITT404: Data Science	**Course Type**	Theory

Course Objectives:

1. To introduce the fundamental concepts of data analytics, including data collection, cleaning, and preprocessing.
2. To explore statistical methods and data mining techniques for data analysis and pattern discovery.
3. To understand the role of machine learning algorithms in predictive analytics and their implementation.
4. To examine advanced data analytics topics such as big data analytics, real-time data processing, and data visualization.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Data Analytics and Preprocessing: Overview of Data Analytics: Definitions, Types, and Applications Data Collection Techniques: Sources, Tools, and Methods Data Cleaning: Handling Missing Data, Outlier Detection, Data Imputation Data Transformation: Normalization, Standardization, Feature Engineering Introduction to R for Data Analytics	9
2	Statistical Methods and Data Mining: Descriptive Statistics: Mean, Median, Mode, Variance, Standard Deviation Inferential Statistics: Hypothesis Testing, Confidence Intervals, p-Values Data Mining Techniques: Classification, Clustering, Association Rule Mining Feature Selection and Dimensionality Reduction: PCA, LDA Applications of Data Mining: Market Basket Analysis, Customer Segmentation	9

3	Predictive Analytics and Machine Learning: Introduction to Predictive Analytics: Concepts and Applications Supervised Learning: Linear Regression, Decision Trees, Random Forests Unsupervised Learning: k-Means Clustering, Hierarchical Clustering Model Evaluation and Optimization: Cross-Validation, Grid Search Case Studies in Predictive Analytics: Financial Forecasting, Health Analytics	**9**
4	Advanced Topics in Data Analytics: Big Data Analytics: Hadoop, Spark, NoSQL Databases Real-Time Data Processing: Streaming Data, Apache Kafka Data Visualization Techniques: Dashboards, Reports, Data Storytelling Ethical Considerations in Data Analytics: Privacy, Security, and Bias Future Trends in Data Analytics: AI-Driven Analytics, Automated Data Science.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Examination (Written)	Evaluate Level Assessment	Analyze Level Assessment	Total
5	15	10	10	40

Evaluate and Analyze Level Assessment [20 Marks]

Students should evaluate and analyze a real-world optimization problem, assess the proposed solutions, provide a conclusion on which solution is most appropriate for the problem, and implement the chosen solution using Python.

Criteria for evaluation:

1. **Problem Definition (K4 - 4 points)**
    a. *Clearly defines the real-world analytical problem.*
    b. *Data is collected from reliable sources and is thoroughly cleaned, transformed, and preprocessed with no errors.*
2. **Problem Analysis (K4 - 4 points)**
    a. *Exploratory Data Analysis is comprehensive, revealing deep insights and*

trends with clear visualizations and interpretations.

   b. *Advanced and appropriate analytical methods are to applied*

3. **Evaluate (K5 - 4 points)**
   a. *Thoroughly evaluate the proposed solutions.*
   b. *Demonstrates strong awareness and adherence to ethical standards in data usage, privacy, and compliance..*
   c. *Considers feasibility, scalability, and practical implications.*
4. **Python Implementation (K5 - 4 points)**
   a. *Select the most feasible solution by implementing the proposed solutions.*
   b. *Successfully translates the chosen solution into Python code.*
   c. *Demonstrates expert use of data analytics tools and technologies*
5. **Conclusion (K4- 2 points, K5 – 2 points)**
   a. *Demonstrates exceptional creativity and innovation in problem-solving.* ***(K4)***
   b. *Results are interpreted accurately with strong, well-supported conclusions and actionable insights.* ***(K5)***

Scoring:

1. **Accomplished (4 points)**: Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points)**: Solid performance with minor areas for improvement.
3. **Developing (2 points)**: Adequate effort but lacks depth or clarity.
4. **Minimal (1 point)**: Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply various techniques for data preprocessing and analytics	**K3**
CO2	Implement statistical and data mining methods for data analysis and pattern discovery.	**K3**
CO3	Utilize machine learning algorithms for predictive analytics in real-world scenarios.	**K3**
CO4	Analyse big data and implement real-time data processing and visualization techniques.	**K3**
CO5	Synthesize and evaluate complex data analytics solutions for large-scale and real-time applications.	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	-	-	2	-	-	-	2
CO2	3	3	3	2	2	-	-	-	2	1	-	2
CO3	3	3	3	2	2	-	-	-	1	-	-	2
CO4	3	3	3	3	2	1	-	3	-	1	-	3
CO5	3	3	3	3	3	2	1	-	2	2	1	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Data Science for Business: What You Need to Know About Data Mining and Data-Analytic Thinking	Foster Provost, Tom Fawcett	O'Reilly Media	1st Edition, 2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython	Wes McKinney	O'Reilly Media	2nd Edition, 2017
2	Data Mining: Concepts and Techniques	Jiawei Han, Micheline Kamber, Jian Pei	Morgan Kaufmann	3rd Edition, 2011
3	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media	2nd Edition, 2019
4	Big Data: Principles and Best Practices of Scalable Real-Time Data Systems	Nathan Marz, James Warren	Manning Publications	1st Edition, 2015

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://www.youtube.com/watch?v=ua-CiDNNj30
2	https://www.youtube.com/watch?v=7eh4d6sabA0
3	https://www.youtube.com/watch?v=7eh4d6sabA0
4	https://www.youtube.com/watch?v=s0dMTAQM4cw

SEMESTER S6

INTERNET OF THINGS

Course Code	**PBITT604**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	**ESE Marks**	40
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Programming Basics	**Course Type**	Theory

Course Objectives:

1. Students will be able to gain a comprehensive understanding of IoT, from its foundational technologies to emerging trends and future directions.
2. Students will be able to develop IoT applications using Arduino and Raspberry Pi, focusing on real-world scenarios and problem-solving.

SYLLABUS

Module No.	Syllabus Description	Contact Hours (44 Hrs)
1	**The Internet of Things Today**: Time for Convergence, Towards the IoT Universe(s), Internet of Things Vision, IoT Strategic Research and Innovation Directions, IoT Applications. Internet of Things and related future Internet technologies, Infrastructure, Networks and Communication, Processes, Data management, Security, Privacy & Trust, Device Level Energy Issues. Current trends in Internet: Internet of Everything, Internet of Thing-Storage, Databases.	**11**
2	**IoT Standardisation**: Status, Requirements, Initiatives and Organisations, M2M Service Layer Standardisation, OGC Sensor Web for IoT, IEEE and IETF, ITU-T. A Common Architectural Approach- The IoT-A Reference Model. IoT Software Platforms: Node-RED, ThingSpeak, Blynk, IoTivity IoT Operating Systems: Contiki, RIOT, TinyOS Cloud Platforms for IoT: AWS IoT, Google Cloud IoT, Microsoft Azure.	**11**
3	**IoT Protocols:** Physical Data Link Layer: IEEE 802.15.4, LoRa, NFC, Zigbee,	**11**

	Bluetooth/Bluetooth Low Energy (BLE), 6LoWPAN Network Layer: IPv4/IPv6, RPL, CoAP Transport Layer: TCP, UDP, DTLS Session & Application Layer: CoAP, MQTT, HTTP/HTTPS, XMPP, AMQP, DDS MQTT Publisher-Subscriber Model using MQTT Broker (Implementation) HTTP - WebSocket communication	
4	**IoT Hardware:** Arduino – Introduction-Making the Sketch. ESP32 System on a Chip (SoC) microcontroller. Simple Digital and Analog Input-Using a Switch, Reading a Keypad, Reading Analog Values Getting Input from Sensors-Detecting Movement, light, motion, distance, vibration, sound and temperature Reading RFID Tags, Getting Location from a GPS Visual Output-Connecting and Using LEDs Physical Output-Controlling the Position of a Servo Remotely Controlling External Devices-Controlling a Digital Camera Connect Arduino to wired and wireless networks. Raspberry Pi – Introduction, Python/Micro python and Raspberry Pi, Arduino and Raspberry Pi, Basic Input and Output, Programming Inputs and Outputs with Python. IoT deployment for Raspberry Pi /Arduino/Equivalent development platforms	**11**

Suggestions on Project Topics:

No.	Project Topics
1	Smart Home Automation System
2	Smart Garden with Environmental Monitoring
3	RFID-based Attendance System
4	GPS-based Vehicle Tracking System
5	IoT-based Weather Station
6	Remote Surveillance System with Camera Control
7	Health Monitoring System
8	Smart Traffic Management System
9	Air Quality Monitoring System
10	Smart Waste Management System
11	IoT-based Smart Parking System
12	Smart Door Lock System
13	IoT-based Personal Safety Device
14	Smart Irrigation System
15	Smart City Infrastructure Monitoring

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions		
Part A	**Part B**	**Total**
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	• Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. • Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	**Bloom's Knowledge Level (KL)**
CO1	Explain the current state and future of the Internet of Things (IoT), covering its key technologies, infrastructure, data management, security, and energy challenges.	**K2**
CO2	Explain IoT standardization, including major organizations and standards, and understand current trends like the Internet of Everything and how storage and databases are used in IoT.	**K2**
CO3	Identify and describe key IoT protocols across different layers.	**K3**
CO4	Apply Arduino and Raspberry Pi to develop and deploy IoT solutions effectively.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	3	2	-	-	-	-	-	-	-
CO2	2	2	3	-	3	-	-	2	2	-	2	2
CO3	3	1	1	3	3	3	-	2	2	-	2	2
CO4	3	2	3	3	3	-	3	3	2	-	2	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems	Dr. Ovidiu Vermesan, Dr. Peter Friess	River Publishers.	1st Edition 2013
2	Arduino Cookbook	Michael Margolis	O'Reilly Media	3rd Edition 2020
3	Getting Started With Raspberry Pi	Matt Richardson, Shawn Wallace	O'Reilly Media	3rd Edition 2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Building The Internet of Things with IPv6 And MIPv6	Daniel Minoli	Wiley	2nd Edition 2019
2	The Internet of Things: Key Applications and Protocols	Olivier Hersent, David Boswarthick, and Omar Elloumi	Wiley	2nd Edition, 2012

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_cs115/
2	https://onlinecourses.swayam2.ac.in/ntr24_ed44/
3	https://onlinecourses.nptel.ac.in/noc24_cs115

Project Based Learning - Course Elements			
L: Lecture (3 Hrs.)	**R: Project (1 Hr.), 2 Faculty Members**		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity		
Sl. No	**Evaluation for**	**Allotted Marks**
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**
    - Clarity and feasibility of the project plan
    - Research and background understanding
    - Defined objectives and methodology
2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
    - Individual contribution to the presentation
    - Effectiveness in answering questions and handling feedback
3. **Involvement in the Project Work and Team Work (3 Marks)**
    - Active participation and individual contribution
    - Teamwork and collaboration
4. **Execution and Implementation (10 Marks)**
    - Adherence to the project timeline and milestones
    - Application of theoretical knowledge and problem-solving
    - Final Result
5. **Final Presentation (5 Marks)**
    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions
6. **Project Quality, Innovation, and Creativity (3 Marks)**
    - Overall quality and technical excellence of the project
    - Innovation and originality in the project
    - Creativity in solutions and approaches

SEMESTER S6

OBJECT-ORIENTED PROGRAMMING IN JAVA

Course Code	**OEITT611**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1.To equip students with the fundamental concepts of Object-Oriented Programming (OOP) using Java.

2.Students will learn how to design, implement, and test Java programs, focusing on encapsulation, inheritance, and polymorphism principles.

3.Students will be equipped with problem-solving, algorithm development, and good programming practices.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Java and OOP:** Overview of programming languages, Setting up the development environment (IDE, JDK), Writing and running a simple Java program **Basics of Java:** Variables, data types, and operators, control structures (conditional control statements, loops), Input/output Basics (reading console input, writing console output)	9
2	**Classes and Objects:** Defining classes and objects, Constructors and instance variables, Methods and method overloading **Encapsulation and Access Modifiers:** Encapsulation principles, Access modifiers (public, private, protected), Getters and setters	9
3	**Inheritance:** Inheritance hierarchy, Method overriding Superclass and subclass relationships **Polymorphism:** Method overloading vs. method overriding, Dynamic method dispatch, Abstract classes and interfaces	9

4	**Exception Handling:** Handling exceptions (try-catch blocks), Custom exceptions, Checked vs. unchecked exceptions. Errors: Understanding common Error types in Java, using Java Error information for root cause analysis and JVM tuning. **Collections and Data Structures:** Lists, sets, and maps, ArrayList, HashSet, HashMap, Iterators	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Demonstrate the ability to set up the Java development environment and write a simple Java program.	**K3**
CO2	Define and instantiate classes and objects in Java, implement constructors and instance variables, and apply encapsulation principles and access modifiers	**K3**
CO3	Implement inheritance hierarchy and polymorphism principles, including method overriding, dynamic method dispatch, and the use of abstract classes and interfaces	**K3**
CO4	Apply exception handling mechanisms and use custom exceptions, and utilize Java collections and data structures like lists, sets, and maps.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2								2
CO2	3	3	3	3								2
CO3	3	3	3	3								2
CO4	3	3	3	3								2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Java: The Complete Reference	Herbert Schildt, Danny Coward	McGraw Hill	13th Edition, 2023
2	Programming with Java	E. Balagurusamy	McGraw Hill	7th Edition, 2023

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Core Java Volume –I Fundamentals	Cay S. Horstmann, Gary Cornell	Prentice Hall	9th Edition, 2013.
2	Java SE 8 for Programmers	Paul Deitel, Harvey Deitel	Pearson	3rd Edition, 2015
3	Beginning Java Programming	Bart Baesens, Aimee Backiel, Seppe vanden Broucke	Wrox,O'Reilly	2015
4	Learn Java Fundamentals – Object-Oriented Programming	Vahe Aslanyan	freeCodeCamp	2023

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_cs47/
2	https://nptel.ac.in/courses/106105191

SEMESTER S6

DATA STRUCTURES USING C++

Course Code	**OEITT612**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To introduce the fundamental concepts of data structures and their implementation using C++.
2. To explore linear data structures such as arrays, linked lists, stacks, and queues.
3. To understand non-linear data structures such as trees and graphs, and their applications.
4. To examine advanced data structures and algorithms for efficient data management and retrieval.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Module 1: Introduction to Data Structures and C++ Basics** Overview of Data Structures: Definition, Importance, and Applications C++ Programming Basics: Pointers, References, Dynamic Memory Allocation Introduction to Abstract Data Types (ADTs) Complexity Analysis: Time and Space Complexity Basic C++ STL Containers: Vectors, Lists, Stacks, and Queues	**9**
2	**Module 2: Linear Data Structures** Arrays and Strings: One-Dimensional and Multidimensional Arrays, String Manipulation Linked Lists: Singly Linked List, Doubly Linked List, Circular Linked List Stacks: Implementation using Arrays and Linked Lists, Applications Queues: Implementation using Arrays and Linked Lists, Circular Queue, Priority Queue	**9**

	C++ STL Implementation of Linear Data Structures	
3	**Module 3: Non-Linear Data Structures** Trees: Binary Trees, Binary Search Trees (BST) Tree Traversal Techniques: Inorder, Preorder, Postorder, Level Order Heaps: Min-Heap, Max-Heap, Heap Sort Graphs: Representation (Adjacency Matrix, Adjacency List), Graph Traversal (DFS, BFS) Applications of Trees and Graphs in Problem Solving	**9**
4	**Advanced Data Structures and Algorithms** Hashing: Hash Functions, Collision Resolution Techniques, Applications Search Algorithms: Binary Search, Interpolation Search, Exponential Search Sorting Algorithms: Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Merge Sort, Heap Sort, Radix Sort C++ STL Implementation of Advanced Data Structures	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply basic data structure concepts to develop simple programs using C++	**K3**
CO2	Implement linear data structures using C++ for various computational tasks.	**K3**
CO3	Utilize non-linear data structures like trees and graphs for complex problem-solving	**K3**
CO4	Analyze and implement advanced data structures and algorithms for optimized performance.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	-	-	-	-	-	1	2	1
CO2	3	3	2	2	1	-	-	-	-	1	2	1
CO3	3	3	3	2	2	1	-	-	-	1	2	2
CO4	3	3	3	3	2	2	1	-	-	1	2	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Data Structures Through C++	Yashwant Kanetkar	BPB	5th Edition,2019
2	Data Structures and Algorithm Analysis in C++	Mark Allen Weiss	Pearson	4th Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	C++ Programming: Principles and Practice Using C++	Bjarne Stroustrup	Addison-Wesley	2nd Edition 2014
2	Data Structures and Algorithms in C++	Adam Drozdek	Cengage Learning	4th Edition, 2012
3	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein	MIT Press	3rd Edition, 2009
4	The Art of Computer Programming, Volumes 1-4A Boxed Set	Donald E. Knuth	Addison-Wesley Professional	Box Set Edition, 2011

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://www.youtube.com/watch?v=ZyRLdtKXW4s
2	https://www.youtube.com/watch?v=MeNaA-eG25k
3	https://www.youtube.com/watch?v=6oL-0TdVy28
4	https://www.youtube.com/watch?v=oaY3m2hVGw8

SEMESTER S6
AI WITH PYTHON

Course Code	OEITT613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide a comprehensive overview of artificial intelligence and its practical applications.
2. To demonstrate proficiency in AI problem definition, algorithm design, and Python implementation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Artificial Intelligence (AI) and Python: Overview of AI and Agents- Definition and History of AI, Narrow AI vs. General AI, AI Applications, Definition and Components of AI Agents, Types of Agents- Simple Reflex, Model Based, Goal Based, and Utility Based Agents, Agent Architectures and Environments. Python for AI- Python basic data structures, Setting Up Python Environment, Key Libraries and Tools- NumPy, Pandas, Matplotlib.	9
2	AI Problems: Definition and formulation of AI problems, State space representation, actions, and goals, Problem characteristics- problem decomposability, solution steps consideration, problem solution predictability, absolute or relative problem solution, state or path solution, role of knowledge in problem solving, human interaction in problem solving. Problem definition and Python implementation of Water Jug Problem, 8-Puzzle Problem & Missionaries and Cannibals Problem.	9
3	AI Search Algorithms: Uninformed Search Algorithm- Breadth First Search and Depth First Search algorithms description and implementation using Python, Informed Search Algorithms- Hill Climbing, A* Search algorithm, and AO* Search algorithm description and implementation using Python, Comparison of informed and uninformed search algorithms.	9

4	Game Playing and Constraint Satisfaction Problem (CSP): Minimax algorithm and Alpha-Beta Pruning, Game Playing Implementation- Tic-Tac-Toe with minimax and alpha-beta pruning, Introduction to CSP- Definitions and basic concepts, Techniques for CSP- Backtracking and Constraint Propagation.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe the evolution and types of AI, outline the role of different AI agents, and set up a Python environment for AI development.	**K2**
CO2	Understand, represent, and solve fundamental AI problems using Python.	**K3**
CO3	Implement various search algorithms used in AI to efficiently solve problems.	**K3**
CO4	Implement game-playing algorithms and constraint satisfaction techniques using Python.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1	2	-	-	-	2	-	-	2
CO2	3	3	3	2	3	2	-	-	2	-	-	2
CO3	3	3	3	2	3	3	-	-	2	-	-	2
CO4	3	3	2	3	3	2	-	2	2	-	-	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	Pearson	4th Edition, 2020
2	Artificial Intelligence: Foundations of Computational Agents	Poole, D. L. and Mackworth, A.K.	Cambridge University Press	2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Artificial Intelligence: A Guide for Thinking Humans	Melanie Mitchell	Penguin Books	1st Edition, 2019
2	Introduction to Artificial Intelligence	Wolfgang Ertel	Springer	1st Edition, 2018
3	Principles of Artificial Intelligence	Nils J. Nilsson	Morgan Kaufmann	2014

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_ge47/preview
2	https://onlinecourses.nptel.ac.in/noc22_cs32/preview
3	https://onlinecourses.nptel.ac.in/noc21_cs79/preview
4	https://onlinecourses.nptel.ac.in/noc22_cs56/preview
5	https://onlinecourses.nptel.ac.in/noc22_cs06/preview

SEMESTER S6

NETWORK SECURITY LAB

Course Code	**PCITL607**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT402: Computer Networks	**Course Type**	Lab

Course Objectives:

1. To understand and implement modern symmetric key algorithms and public key infrastructure in network security.
2. To explore and apply digital signatures and cryptographic hash functions in secure communication.
3. To study and implement intrusion detection and prevention systems for securing networks.
4. To configure and test firewalls, network security protocols, and cryptographic protocols.

Experiment No.	Experiments
1	Implementation of AES and DES algorithms for data encryption and decryption.
2	Setting up and configuring a public key infrastructure (PKI) for secure communication.
3	Key management using RSA algorithm and certificate generation in a simulated environment.
4	Implementation of digital signatures using RSA and DSA algorithms.
5	Application of cryptographic hash functions (SHA-256, MD5) for data integrity verification.
6	Secure message transmission using digital signatures and hash functions.
7	Configuration and deployment of Snort as an Intrusion Detection System (IDS).
8	Implementation of an Intrusion Prevention System (IPS) using open-source tools.

9	Analysis of network traffic and identification of malicious activities using IDS/IPS.
10	Configuration of a network firewall using iptables/pfSense.
11	Implementation of VPN protocols (IPSec, SSL/TLS) for secure communication.
12	Testing and validation of cryptographic protocols (SSL/TLS) in securing web transactions.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record:*** *Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement modern symmetric key algorithms and configure public key infrastructure to secure data transmission.	**K3**
CO2	Use digital signatures and cryptographic hash functions for secure communication and data integrity.	**K3**
CO3	Apply and evaluate intrusion detection and prevention systems to protect network infrastructures.	**K3**
CO4	Configure and test firewalls and network security protocols to safeguard networks.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	2	-	-	1	1	-	-	2
CO2	3	3	3	2	2	-	-	1	1	-	-	2
CO3	3	3	3	3	2	-	-	-	2	-	-	3
CO4	3	3	3	3	2	-	-	-	2	1	-	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cryptography and Network Security: Principles and Practice	William Stallings	Pearson Education Press	7th Edition, 2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Applied Cryptography: Protocols, Algorithms, and Source Code in C	Bruce Schneier	Wiley	20th Anniversary Edition, 2015
2	Network Security Essentials: Applications and Standards	William Stallings	Pearson Education	6th Edition, 2017

3	Network Security with OpenSSL	John Viega, Matt Messier, Pravir Chandra	O'Reilly Media	1st Edition, 2002
4	Firewalls and Internet Security: Repelling the Wily Hacker	William R. Cheswick, Steven M. Bellovin, Aviel D. Rubin	Addison-Wesley	2nd Edition, 2003

Video Links (NPTEL, SWAYAM…)	
Link No.	**Link ID**
1	https://www.youtube.com/watch?v=ZD71JeX4VkI
2	https://www.youtube.com/watch?v=3QnD2c4Xovk
3	https://www.youtube.com/watch?v=jvf4eCW1XI4
4	https://www.youtube.com/watch?v=8yKpQSHpYP0

Continuous Assessment (25 Marks):

1. **Preparation and Pre-Lab Work (7 Marks)**
    - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
    - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.
2. **Conduct of Experiments (7 Marks)**
    - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
    - Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
    - Teamwork: Collaboration and participation in group experiments.
3. **Lab Reports and Record Keeping (6 Marks)**
    - Quality of Reports: Clarity, completeness, and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
    - Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**
   - Oral Examination: Ability to explain the experiment, results, and underlying principles during a viva voce session.

Final Marks Averaging: *The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.*

Evaluation Pattern for End Semester Examination (50 Marks):

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**
   - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
   - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
   - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
   - Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**
   - Setup and Execution: Proper setup and accurate execution of the experiment or programming task.
3. **Result with Valid Inference/Quality of Output (10 Marks)**
   - Accuracy of Results: Precision and correctness of the obtained results.
   - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.
4. **Viva Voce (10 Marks)**
   - Ability to explain the experiment, procedure results and answer related question.
   - Proficiency in answering questions related to theoretical and practical aspects of the subject.
5. **Record (5 Marks)**
   - Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

INFORMATION TECHNOLOGY